Review according to the instructions in the document.

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User prompt: Review according to the instructions in the document.

Response: **Verdict:** While the whitepaper is unusually candid about its engineering defects and operational compromises, its load-bearing mathematical scaffolding breaks at three critical points that invalidate its core statistical and power claims. First, the holdout power derivation in Appendix S4a uses an unconstrained Kelly approximation ($\log K \simeq nS_{\text{step}}^2/2$) that violates the mandatory Ville supermartingale cap $\lambda_t \leq 1/2$ by more than an order of magnitude at the paper's own baseline volatility ($\sigma = 2\%$ per bar); under the cap, reaching $p = 0.05$ in $n = 546$ steps requires an annualized Sharpe ratio of **25.9**, not the claimed **4.9**, which directly explains why Stage 2 failed with zero power. Second, capping reported step returns at -1 (Theorem 6) gives zero-mean pre-absorption null strategies with leverage $\Lambda = 4$ a strictly positive expected reported return ($\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] > 0$), corrupting every search-period test (DSR, SPA, Romano–Wolf) and allowing strategies that go completely bankrupt during the holdout to pass the running-maximum betting test ($p = 1/\max_t K_t \leq 0.05$)—while the frozen pilot's unpatched "bankruptcy resurrection" defect breaks Appendix S3's proof that reported and pre-absorption running maxima agree. Third, Appendix E2's proof of the pre-absorption lower bound $r'_t \geq -2$ relies on a non-causal settlement order (Appendix D1) that credits end-of-bar closing funding qcf_4 to positions already liquidated intrabar at the wick (contradicting Appendix E1); under chronological intrabar liquidation, r'_t drops to $-2.072 < -2$ (and to $-2.16 < -2$ after end-of-bar funding erodes equity below the $\Lambda = 4$ mark-gross bound), breaking the nonnegativity required by Ville's inequality.

Finding 1 — Unconstrained Kelly Power Formula Violates the Mandatory Ville Cap $\lambda_t \leq 1/2$ by $10.5\times$

- **Severity:** Critical
- **Location:** Appendix S4a (`\label{app:power}`), Section 7 (`\label{sec:gate}`), and Section 12 (`\label{sec:findings}`, "Calibration and sensitivity (stage 2)"):

  > "At a saturated half-cap, $\log K_n \simeq n(\mu/2 - \mathrm{E}r^2/8)$, so ignoring the quadratic cost, one finalist needs $\mu \simeq 2\log 20/n$ to reach $p = 0.05$: about 1.66%, 1.10%, 0.55% and 0.11% per step at $n = 360, 546, 1,080, 5,400$. Holm raises that hurdle. An unconstrained second-order Kelly approximation gives $\log K \simeq nS_{\text{step}}^2/2$, hence annualized $S \simeq \quad 2\log 20\,(6 \cdot 365)/546 \simeq 4.9$ at 546 steps."

- **The Claim:** Under a second-order Kelly approximation for the holdout betting test, a single finalist needs an annualized Sharpe ratio of approximately $S \simeq 4.9$ ($S_{\text{step}} \simeq 0.1048$ per 4-hour bar) to reach $p = 0.05$ ($\log K_n \geq \log 20 \approx 2.9957$) over $n = 546$ steps (91 days).
- **Why It Is Wrong (with Recomputation):**

  1. In Appendix S3 (`\label{app:bet}`), the second-order Taylor expansion of expected log-wealth for a constant bet λ is $\mathbb{E}[\log(1 + \lambda r)] \simeq \lambda\mu - \frac{1}{2}\lambda^2\mathbb{E}[r^2]$. Obtaining $\mathbb{E}[\log K_n] \simeq \frac{1}{2}nS_{\text{step}}^2$ requires setting λ equal to the **unconstrained** oracle Kelly fraction:

  $$\lambda^* = \frac{\mu}{\mathbb{E}[r^2]} \simeq \frac{\mu}{\sigma^2} = \frac{S_{\text{step}}}{\sigma}.$$

  2. However, Appendix S3 proves that Ville's supermartingale inequality under the lower bound $r'_t \geq -L = -2$ strictly requires capping the betting fraction at $\lambda_t \leq 1/L = 1/2$ ("Ville's nonnegativity requires $\lambda_t \leq 1/L$ under $r' \geq -L$, so any later bet must keep $\lambda_t \leq 1/L$").

  3. The unconstrained fraction $\lambda^* = S_{\text{step}}/\sigma$ stays within the cap $\lambda_t \leq 1/2$ if and only if per-step volatility satisfies $\sigma \geq 2S_{\text{step}}$. At $n = 546$, $S_{\text{step}} = \quad 2\log 20/546 = 0.10475$, so the unconstrained approximation is valid only if per-step volatility is at least $\sigma \geq 2(0.10475) = 20.95\%$ **per 4-hour bar**—which corresponds to $20.95\% \times \quad 6 \cdot 365 = \mathbf{980\%}$ **annualized volatility**!

  4. At the paper's own baseline crypto bar volatility of $\sigma = 2\%$ per 4-hour bar (used in N1 and C1.7, Appendices D1 and K1), a strategy with annualized Sharpe $S = 4.9$ ($S_{\text{step}} = 0.10475$, mean per-step return $\mu = S_{\text{step}}\sigma = 0.002095 = 0.2095\%$) has an unconstrained Kelly fraction of:

  $$\lambda^* = \frac{0.10475}{0.02} = \mathbf{5.2375},$$

  which exceeds the mandatory cap $\lambda_t \leq 1/2$ by a factor of $10.5\times$.

  5. Because the actual bet is capped at $\lambda_t \leq 1/2$ (and starts at $\lambda_1 = 0$ with noisy plug-in estimates $\min(1/2, \max(0, S_{t-1}/Q_{t-1}))$), the actual expected log-wealth at $n = 546$, $\sigma = 0.02$, and $S_{\text{ann}} = 4.9$ is bounded above by the saturated half-cap drift:

  $$\mathbb{E}[\log K_{546}] \leq 546\left(\frac{\mu}{2} - \frac{\sigma^2 + \mu^2}{8}\right) = 546\left(\frac{0.002095}{2} - \frac{0.0004044}{8}\right) = 546(0.0010475 - 0.0000505) = \mathbf{0.5443} \ll \log($$

This yields $K_{546} \approx e^{0.5443} = 1.723$, or $p \approx \mathbf{0.580}$—nowhere near the $p = 0.05$ threshold!

6. Recomputing the true required annualized Sharpe at $\sigma = 0.02$ and $n = 546$ under the $\lambda_t \le 1/2$ cap: solving $546(\mu/2 - (\sigma^2 + \mu^2)/8) = \log 20$ gives $\mu = 0.01109 = 1.11\%$ per 4-hour bar (matching S4a's first sentence!), which at $\sigma = 2\%$ per bar is a per-step Sharpe of $S_{\text{step}} = 0.01109/0.02 = 0.5545$, or an **annualized Sharpe ratio of**:

$$S_{\text{ann}} = 0.5545 \quad 6 \cdot 365 = \mathbf{25.95} \quad (\text{not } 4.9).$$

- **Which Result or Decision It Changes:** It invalidates the $S \simeq 4.9$ power benchmark in Appendix S4a, directly explains why Stage 2's power check C2.6 failed with **0/100 promotions** on planted Sharpes of 1, 2, 2.5, and 3 in `N2-s3` (and why even `N2-long` at $n = 5,400$ steps with $\mu = 0.2\%$ per step had only 5/100 promotions, requiring a human waiver to exit Stage 2), and proves that the pilot's 91-day forward holdout ($n = 546$) has essentially zero power for any strategy with realistic volatility ($\sigma \approx 2\%$ per bar) and annualized Sharpe < 20.

Finding 2 — Capping Reported Losses at -1 Breaks the Null for Search-Period Tests, Promotes Bankrupt Paths on the Holdout, and Fails Under the Pilot's Bankruptcy Resurrection Defect

- **Severity:** Critical
- **Location:** Section 6 (`\label{sec:verified}`, Theorem 6), Section 7 (`\label{sec:gate}`, Steps 2–4 & Step 7), Section 11 (`\label{sec:roadmap}`, Stage 2), Section 12 (`\label{sec:findings}`, "Adversarial findings remain findings"), Appendix S3 (`\label{app:bet}`), and Appendix D1 (`\label{app:nulls}`):

  > "6. **Bounded returns.** Losses are capped at account equity, so every reported per-step return is at least -1." "Bankruptcy's reported -1 and the pre-absorption terminal loss cannot create a new maximum, and later reported returns are zero. Thus their running maxima agree, and Ville [39] gives $p = \min(1, 1/\max K)$... The pilot's reported floor $B_{\text{rep}} = 1$ follows from absorption and zero subsequent equity (I10/L6). It does not imply this conditional null, because $\mathbb{E}[\max(r', -B_{\text{rep}}) \mid \mathcal{F}] = \mathbb{E}[r' \mid \mathcal{F}] + \mathbb{E}[(-B_{\text{rep}} - r')_+ \mid \mathcal{F}]$." "The defects are future-dependent delisting, reopening after stop-out, unbudgeted holds, pre-fill sizing equity, bankruptcy resurrection..."

- **The Claim:** (a) In the frozen pilot and corrected design, the holdout test's running maximum $\max_t K_t$ computed on reported returns $r_t = \max(r'_t, -1)$ equals the pre-absorption running maximum $\max_t K'_t$, preserving conditional Type I error control $P(p_i \le u \mid \mathcal{F}_{\text{search}}, H_{0,i}) \le u$; (b) requiring positive holdout return alongside $p_i \le \alpha_{\text{Holm}}$ prevents promoting losers; and (c) the null models have "zero expected net-of-cost return for every strategy expressible in the DSL" for the search-period gate (DSR, Hansen's SPA, Romano–Wolf) and C2.1's probes.

- **Why It Is Wrong (with Concrete Counterexamples):**

  1. **Search-period tests (DSR, SPA, Romano–Wolf) and C2.1 probes operate on arithmetic means of reported returns $r_t = \max(r'_t, -1)$, where the null expectation is strictly positive:** Unlike Ville's multiplicative wealth process, DSR, Hansen's SPA, Romano–Wolf, and C2.1's per-step z-test/regression probes compute the sample mean $\bar{r} = \frac{1}{T}\sum_{t=1}^{T} r_t$ of reported step returns. Under a leveraged strategy ($\Lambda = 4$) where pre-absorption return $r'_t \in [-2, \infty)$ satisfies $\mathbb{E}[r'_t \mid \mathcal{F}_{t-1}] = 0$, capping the bankruptcy return at -1 makes $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = \mathbb{E}[r'_t \mid \mathcal{F}_{t-1}] + \mathbb{E}[(-1 - r'_t)_+ \mid \mathcal{F}_{t-1}] > 0$. Even without any venue shortfall ($r'_t \ge -1$), a strategy that goes **100% bankrupt** has a strictly positive arithmetic mean of reported step returns: if a strategy earns $r_1 = +1.5$ on step 1 (equity $\$10,000 \to \$25,000$), loses 100% of its equity on step 2 ($r_2 = -1.0$, equity $\$25,000 \to \$0$), and has $r_t = 0$ for $t = 3, \ldots, T$, its actual dollar return is -100% **(total ruin)**, yet its mean per-step return is $\bar{r} = (+1.5 - 1.0)/T = +0.5/T > 0$ with a positive Sharpe ratio $\hat{S} > 0$!

  2. **The holdout test's running maximum $p = \min(1, 1/\max_{1 \le t \le T} K_t)$ rejects H_0 on strategies that go bankrupt during the holdout:** Suppose a finalist earns $r_t = +0.30$ on steps $t = 1, \ldots, 23$ of the holdout, and then goes **completely bankrupt** on step 24 ($r'_{24} = -1.5 \implies r_{24} = -1.0$, terminal equity $e_T = \$0$). With $\lambda_1 = 0$ and $\lambda_t = 1/2$ for $t = 2, \ldots, 23$, the wealth at step 23 reaches $K_{23} = (1.15)^{22} = \mathbf{21.645} > \mathbf{20}$. On step 24 (bankruptcy), $K_{24} = 21.645(1 - 0.5) = 10.822$, and $K_t = 10.822$ thereafter. Because $p = 1/\max_{1 \le t \le T} K_t = 1/21.645 = \mathbf{0.0462} < \mathbf{0.05}$, the holdout test **certifies a strategy that went 100% bankrupt on the holdout**! If Step 7 evaluates holdout return as $\sum_{t=1}^{T} r_t = 23(0.30) - 1.0 = +5.90 > 0$, the bankrupt strategy is promoted to paper trading; even if Step 7 checks terminal dollar equity $e_T > e_0$, a strategy that peaks at $K_{23} = 21.645$ and then loses 99.9% of its gains so that $e_T = 1.0001 e_0$ is promoted with $p = 0.0462$.

  3. **The frozen pilot's "bankruptcy resurrection" defect invalidates Appendix S3's running-maximum equivalence proof:** Appendix S3 proves $\max_t K_t^{\text{rep}} = \max_t K'_t$ only under the premise that "later reported returns are zero" after bankruptcy. Yet Section 12 explicitly admits that in the frozen pilot (used for all judged runs in Stages 1–4), accounts suffer from **bankruptcy resurrection** ("Bankrupt accounts and their pending orders cannot revive within a reporting window" is an unjudged post-pilot fix in K1). When a bankrupt account has $r'_\tau = -2$ capped at $r_\tau = -1$, its pre-absorption wealth is $K'_\tau = K_{\tau-1}(1 - 2\lambda_\tau) = 0$ (at $\lambda_\tau = 1/2$), whereas

its reported wealth is $K_\tau^{\text{rep}} = K_{\tau-1}(1-\lambda_\tau) = \frac{1}{2}K_{\tau-1} > 0$. If the account resurrects at $t > \tau$ and achieves positive returns, K_t^{rep} continues compounding from $\frac{1}{2}K_{\tau-1} > 0$ and can exceed $\max_{s<\tau} K'_s$, creating false rejections under H_0.

- **Which Result or Decision It Changes:** It invalidates the theorem that reported-return holdout p-values equal pre-absorption p-values in the frozen pilot, breaks the zero-mean null property for arithmetic-mean tests (DSR, SPA, Romano–Wolf, C2.1) whenever trajectories hit -1, and shows the holdout promotion rule fails to guard against catastrophic post-peak drawdowns or bankruptcy.

Finding 3 — Appendix E2's Pre-Absorption Bound $r'_t \geq -2$ Fails ($r'_t = -2.072$ and -2.16) Under Chronological Settlement and Contradicts Appendix E1

- **Severity:** Critical
- **Location:** Appendix E2 (`\label{app:loss}`), Appendix D1 (`\label{app:nulls}`), and Appendix E1 (`\label{app:execution}`):

  "Settlement precedes fills at equal instants, so that the position that actually held the hour pays for it." "With hourly $|f| \leq 0.04$, $c/o \leq (1 + 3 \cdot 0.04 + 0.3)/(1 - 0.04) < 1.48$. An adverse closeout loses at most $0.3 + 0.1 \cdot 1.48 < 0.448$ before closeout cost when the bar moves against the position, or $0.1 + 3 \cdot 0.04 + 0.04 \cdot 1.48 < 0.28$ when it moves for it. Closeout cost adds at most $1.1 \cdot 1.48\kappa$, so $m < 0.46 \leq 1/2 - 2\kappa$... $\Lambda = 4,\ m \leq \frac{1}{2} - 2\kappa \implies r' \geq -2$." "Under the simulator's settlement order, a predictable position q held through a bar earns $q(c-o) - qo(f_1+f_2+f_3) - qcf_4$... Long liquidation adds $q(\text{low} - c) \leq 0$, short liquidation adds $q(\text{high} - c) \leq 0$."

- **The Claim:** In the corrected gapless synthetic market ($|\varepsilon| \leq 0.3$, $|u| \leq 0.1$, $|f| \leq 0.04$, $\kappa \leq 0.003$, $\Lambda = 4$), pre-fill and post-fill mark gross satisfy $G_0, G_1 \leq \Lambda e_{\text{prev}}$, single-bar carrying loss per unit notional satisfies $m < 0.46 \leq \frac{1}{2} - 2\kappa = 0.494$, and every step's pre-absorption return satisfies $r'_t \geq -2$.

- **Why It Is Wrong (with Recomputation and Counterexample):** In a 4-hour bar $[B, B+4\text{h})$, funding f_1, f_2, f_3 settles at $B+1\text{h}, B+2\text{h}, B+3\text{h}$ (at *o*), while f_4 settles at the closing boundary $B+4\text{h}$ (at *c*). Intrabar liquidation (stop-out) occurs at the intrabar wick (low or high).

  1. **Under chronological / E1-compliant settlement (intrabar liquidation occurs before the $B+4\text{h}$ close *c*, so a liquidated position does not settle $f_4 c$):** Consider a unit-notional short position ($q = -1/o$) on a bar with $f_1 = f_2 = f_3 = f_4 = +0.04$, $\varepsilon = +0.3$, and wick multiplier $u = 0.1$. Then $c/o = (1 + 3(0.04) + 0.3)/(1 - 0.04) = 1.42/0.96 = 1.479167$, and the adverse wick is $\text{high}/o = 1.1(c/o) = 1.627083$. The short is liquidated intrabar at high, losing $(\text{high} - o)/o = 0.627083$ on price while receiving $f_1 + f_2 + f_3 = 0.12$ in funding for the first 3 hours. Its loss before closeout cost is:

  $$\frac{\text{high} - o}{o} - (f_1 + f_2 + f_3) = 1.627083 - 1 - 0.12 = \mathbf{0.507083} > 0.448.$$

  Adding the closeout cost $1.1(c/o)\kappa = 1.627083 \times 0.003 = 0.004881$ gives:

  $$m = 0.507083 + 0.004881 = \mathbf{0.51196} > 0.494 = \frac{1}{2} - 2\kappa,$$

  which after a reversal from $G_0 = 4e_{\text{prev}}$ to $G_1 = 4e_{\text{prev}}$ yields:

  $$r'_t = -\Lambda(2\kappa + m) = -4(0.006 + 0.51196) = -\mathbf{2.0718} < -\mathbf{2}.$$

  Furthermore, under chronological settlement, a long position that barely survives the intrabar stop-out check with $G_{\text{adv}} = \Lambda e_{\text{adv}}$ (with $u = 0$, so $\text{low} = c$) subsequently pays the 4th-hour funding $f_4 c = 0.04 G_0$ at $B+4\text{h}$. This reduces equity to $e_{\text{prev}} = (1 - 0.04\Lambda)e_{\text{adv}} = 0.84 e_{\text{adv}}$, leaving mark gross at:

  $$G_0 = \frac{\Lambda}{1 - 0.04\Lambda} e_{\text{prev}} = \frac{4}{0.84} e_{\text{prev}} = \mathbf{4.7619}\, \mathbf{e}_{\text{prev}} > \mathbf{4}\, \mathbf{e}_{\text{prev}},$$

  violating E2's claim that $G_0 \leq \Lambda e_{\text{prev}}$! If the strategy then emits an undefined weight (which K1 states holds the existing position) or a tiny sign-preserving reduction (which E2 states bypasses margin admission), $G_1 = 4.7619 e_{\text{prev}}$ is carried into the next bar, where an adverse closeout produces $r'_{t+1} \approx -4.7619 \times 0.4528 = -\mathbf{2.156} < -\mathbf{2}$.

  2. **Under D1's non-causal settlement order ($B+4\text{h}$ funding qcf_4 is settled at the future closing price c before intrabar liquidation):** In E2's derivation, the only reason the liquidated short loses $0.3 + 0.1(1.4792) = 0.4479$ instead of 0.5071 is that D1 credits the short with $+f_4(c/o) = +0.0592$ of end-of-bar funding *at the closing price c* even though the position was liquidated intrabar at high. This directly violates E1's invariant ("so that the position that actually held the hour pays for it") and introduces intrabar lookahead into the stop-out check (making intrabar liquidation equity depend on the end-of-bar close *c*).

- **Which Result or Decision It Changes:** Under any causal intrabar liquidation model, r'_t can fall below -2, which makes $1 + \lambda_t r'_t < 0$ at $\lambda_t = 1/2$, breaking the nonnegativity required by Ville's theorem in Appendix S3 and invalidating the holdout test's Type I error guarantee.

Finding 4 — Stage 4 Clopper–Pearson Bounds (0.133 and 0.198) Pool Across 7 Zero-Evaluation Runs, 1 Unaudited Run, and Redesigned Versions

- **Severity:** High
- **Location:** Section 12 (`\label{sec:findings}`, "Swarm smoke failures (stage 4)") and Appendix S4 (`\label{app:calibration}`):

  > "V2 failed with two counted campaigns out of ten. In each, an agent ran `/work/mm-agent` instead of the permitted relative command, received no evaluation and stopped. One campaign also exposed a crash in reporting a missing log. No false discovery was observed, but neither version passed. V3 passed as written: ten campaigns, none promoted, one unresolved (N2-8, whose agent attempted the unpermitted `Edit` tool, refused by the CLI with no breach, so its audit does not hold and the frozen rule leaves the campaign unresolved rather than established; it would not change the verdict). Pooled over the three versions, 22 campaigns started and 21 are known non-promotions, an upper bound of 0.133 at 95%; one campaign of v2 crashed without a gate outcome, and counted as a promotion the bound is 0.198."

- **The Claim:** Pooling across Stage 4 versions v1, v2, and v3 yields $n = 21$ valid non-promoting trials (and $x = 1$ of $n = 22$ if the v2 crash is counted as a promotion), establishing a 95% one-sided Clopper–Pearson upper bound of 0.133 (or 0.198) on the swarm gate's false-promotion rate.
- **Why It Is Wrong (with Recomputation):** The reported numbers $0.133 = 1 - 0.05^{1/21}$ ($x = 0, n = 21$) and 0.198 (solving $(1 - q)^{21}(1 + 21q) = 0.05$ for $x = 1, n = 22$) require all 21 non-crashing campaigns across v1 (2), v2 (9), and v3 (10) to be i.i.d. $\mathrm{Bernoulli}(q)$ trials of the swarm gate procedure. This is false for three reasons:

  1. **Seven campaigns in v2 made zero evaluations ($P(\text{promotion}) \equiv 0$):** In 7 of v2's 9 non-crashing campaigns, the agent invoked the wrong binary path (`/work/mm-agent`), "received no evaluation and stopped." Because the gate only promotes strategies that were evaluated and filled at least once (Appendix S5), a campaign with zero evaluations has promotion probability **identically zero** ($p_i = 0 \neq q$). Counting 7 structurally impossible promotions as $\mathrm{Bernoulli}(q)$ trials inflates n and fabricates statistical confidence out of broken CLI invocations. Moreover, this directly contradicts the paper's own rule in the immediately preceding paragraph ("Single agent (stage 3)"): *"V2 made no evaluations because workspace writes were denied, so it established no procedure despite its numerical count."*
  2. **One campaign in v3 (N2-8) failed its audit and was left "unresolved rather than established,"** yet is counted among the 21 "known non-promotions."
  3. **Optional stopping and redesign across versions:** v1 was halted after $n = 2$ runs failed the 200-evaluation floor, and v2 was halted after 10 runs failed, violating the fixed-n binomial assumption of Clopper–Pearson. **Recomputation:** Across all three versions, the number of campaigns that actually evaluated strategies and established an audited gate outcome is at most $2\ (\text{v1}) + 2\ (\text{v2 counted}) + 9\ (\text{v3 established}) = \mathbf{13}$ (or $\mathbf{9}$ in the passing v3 version alone):
     - For $x = 0, n = 13$: $q_{0.95} = 1 - 0.05^{1/13} = \mathbf{0.206}$ (not 0.133).
     - Counting the v2 crash as a promotion ($x = 1, n = 14$): solving $(1 - q)^{13}(1 + 13q) = 0.05$ gives $q_{0.95} = \mathbf{0.300}$ (not 0.198).
     - For the passing v3 run's 9 established campaigns alone ($x = 0, n = 9$): $q_{0.95} = 1 - 0.05^{1/9} = \mathbf{0.283}$, which exceeds Appendix S4's stated zero-promotion smoke bound for 10 runs ($1 - 0.05^{1/10} = \mathbf{0.259} \approx 26\%$).
- **Which Result or Decision It Changes:** It invalidates the pooled 0.133 and 0.198 confidence bounds in Section 12 and shows that v3's 9 established campaigns do not meet the 10-run zero-promotion bound (25.9%) cited in Appendix S4.

CONTINUE FOR PART 2

User prompt: continue

Response: Finding 5 — The Mechanism Check Has Up to 99.9% Pass Rate Under the Complete Null Because $\mathrm{Cov}(\bar{r}_A - \bar{r}_B, \bar{r}) = (\sigma_A^2 - \sigma_B^2)/n > 0$

- **Severity:** High
- **Location:** Section 7 (`\label{sec:gate}`, Step 5), Section 8 (`\label{sec:mechanism}`), and Appendix S5 (`\label{app:selection}`):

  > "5. **Mechanism check.** A candidate whose pre-registered mechanism prediction fails is removed from the finalists. There is no override by a higher statistical threshold." "Ideally they are tested on data not used to find the strategy, since predictions

checked on the same sample are not independent of the PnL result… A liquidation-cascade strategy should earn more when positions are crowded near liquidation and less when funding has normalized." "The mechanism's positive sample-mean difference matches its stated sample claim. The 10--90% exposure shares are an underived choice to keep the claim from being vacuous."

- **The Claim:** Requiring a strategy's conditional mechanism prediction to hold (specifically, partitioning active exposure into regime A and regime B with exposure shares $w_A, w_B \in [0.10, 0.90]$ and requiring the sample-mean difference $\bar{r}_A - \bar{r}_B > 0$) acts as an empirical filter that removes spurious search-period survivors.
- **Why It Is Wrong (with Derivation and Counterexample):**
  1. In the implementation (Appendix S5), the mechanism check is evaluated **on the same search-period sample** used to select the strategy, and tests only the sign of the unstandardized difference $D = \bar{r}_A - \bar{r}_B > 0$. Even for an unselected strategy under a homoskedastic null ($\sigma_A = \sigma_B$), a zero-mean symmetric difference satisfies $P(D > 0 \mid H_0) = 0.50$, passing half of all noise strategies.
  2. Worse, in Step 5 of the gate, the mechanism check is applied **only after** a strategy has been selected for having a large positive overall active sample mean $S = w_A\bar{r}_A + w_B\bar{r}_B \gg 0$ (having passed DSR ≥ 0.95, SPA, and Romano–Wolf). Suppose a strategy has zero true edge in every period ($\mathrm{E}[r_t \mid \mathcal{F}_{t-1}] = 0$), with active bars partitioned into $n_A = w_A n$ bars in regime A (variance σ_A^2) and $n_B = w_B n$ bars in regime B (variance σ_B^2). Under the null, the covariance between the mechanism test statistic $D = \bar{r}_A - \bar{r}_B$ and the overall active sample mean $S = w_A\bar{r}_A + w_B\bar{r}_B$ is:
$$\mathrm{Cov}(D, S) = \mathrm{Cov}(\bar{r}_A - \bar{r}_B,\ w_A\bar{r}_A + w_B\bar{r}_B) = w_A\frac{\sigma_A^2}{n_A} - w_B\frac{\sigma_B^2}{n_B} = \frac{\sigma_A^2 - \sigma_B^2}{n}.$$
     Notice that the exposure shares $w_A = n_A/n$ and $w_B = n_B/n$ **cancel out completely**: the $[0.10, 0.90]$ exposure-share rule in Appendix S5 has zero effect on the covariance $\mathrm{Cov}(D, S)$.
  3. Whenever the mechanism predicts higher returns in the more volatile regime ($\sigma_A > \sigma_B$—as in Section 8's flagship example where regime A is "crowded near liquidation" and regime B is "when funding has normalized"), D and S are **strictly positively correlated under the complete null**.
  4. **Concrete numerical example:** Take $w_A = 0.20, w_B = 0.80$, and $\sigma_A = 3\sigma_B$ ($\sigma_A^2 = 9\sigma_B^2$) under an independent Gaussian null. Then:
$$\mathrm{Var}(D) = \left(\frac{9}{0.20} + \frac{1}{0.80}\right)\frac{\sigma_B^2}{n} = 46.25\frac{\sigma_B^2}{n}, \qquad \mathrm{Var}(S) = (0.20(9) + 0.80(1))\frac{\sigma_B^2}{n} = 2.60\frac{\sigma_B^2}{n},$$
$$\rho(D, S) = \frac{9 - 1}{46.25 \times 2.60} = \frac{8}{10.9659} = +\mathbf{0.7295}.$$
     Conditional on a pure-noise strategy passing the search screen with a t-statistic of $z_S = S/\quad \mathrm{Var}(S) = 3.0$, the conditional distribution of $Z_D = D/\quad \mathrm{Var}(D)$ under H_0 is $\mathcal{N}(\rho z_S, 1 - \rho^2) = \mathcal{N}(2.1886, 0.4678)$, so the probability that the **null strategy passes the mechanism check** is:
$$P(\bar{r}_A - \bar{r}_B > 0 \mid z_S = 3.0, H_0) = \Phi\left(\frac{2.1886}{0.4678}\right) = \Phi(3.200) = \mathbf{99.93\%}.$$
- **Which Result or Decision It Changes:** It shows that Step 5 of the statistical gate provides virtually zero filtering power (99.9% pass rate under H_0) for any mechanism that designates higher-volatility states as the active regime on the search sample, and proves that the 10–90% exposure-share rule fails to prevent this degeneracy.

Finding 6 — C2.5's Hoeffding Bound on Unconditional FDP ($\bar{X} \leq 0.0776$) Is Vacuous When Power Is Low ($\leq 7\%$), Passing Gates Whose Promotions Are 100% False

- **Severity:** High
- **Location:** Section 11 (`\label{sec:roadmap}`, Stage 2), Section 12 (`\label{sec:findings}`, "Calibration and sensitivity (stage 2)"), and Appendix S4 (`\label{app:calibration}`):

  "For campaign FDP $X \in [0, 1]$, Hoeffding gives $\mathrm{E}X \leq \bar{X} + \quad \log(1/\delta)/(2n)$ with confidence $1 - \delta$. C2.5 chooses $n = 100$, $\delta = 0.05$, target 0.20, giving $\bar{X} \leq 0.20 - \quad \log 20/200 \simeq 0.0776$. C9.5's union bound over M mixed cells gives penalty $\quad \log(M/0.05)/(2n)$…" "After a revised reference scale and an independent implementation, the r2 exit record reports C2.1--C2.5 met. Every complete-null cell had zero promotions in 300 campaigns, and every judged Sharpe mixed cell had zero FDR… C2.6 failed in both rounds, as written and amended: r2 N2-s3 power was zero and N2-long was 5/100; stage 2 exited only by the human's sensitivity waiver, not a pass."

- **The Claim:** Observing sample mean $\bar{X} = \frac{1}{100}\sum_{i=1}^{100} X_i \le 0.0776$ across $n = 100$ mixed-signal-and-noise campaigns certifies at 95% confidence that the gate controls the false discovery rate among promotions below 0.20, and Stage 2 met C2.5 across all judged mixed cells.
- **Why It Is Wrong (with Recomputation and Counterexample):**
  1. **Unconditional $\mathrm{FDR} = \mathrm{E}[V/\max(R,1)]$ vs. Positive $\mathrm{pFDR} = \mathrm{E}[V/R \mid R > 0]$:** In campaign i, let R_i be the total number of promoted finalists and $V_i \le R_i$ be the number of false promotions. Because $X_i = \mathrm{FDP}_i = \frac{V_i}{\max(R_i,1)}$ is defined as 0 whenever a campaign promotes nothing ($R_i = 0$), X_i is deterministically bounded above by the promotion indicator:
  $$X_i = \frac{V_i}{\max(R_i,1)} \le \mathbf{1}\{R_i > 0\} \implies \bar{X} \le \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{R_i > 0\}.$$
  2. Consequently, whenever a gate promotes in at most 7 out of 100 mixed campaigns—as occurred throughout Stage 2 (`N2-s3` promoted in **0/100** campaigns; `N2-long` promoted in **5/100** campaigns)—the sample average $\bar{X}$ is **at most $0.07 \le 0.0776$ even if 100% of all promoted strategies are false discoveries ($V_i = R_i$ whenever $R_i > 0$, i.e., $\mathrm{pFDR} = 100\%$)**!
  3. **Concrete counterexample:** Suppose in $n = 100$ mixed campaigns, the gate completely ignores the planted real signals and instead promotes a pure-noise strategy in 5 campaigns ($V_i = R_i = 1$ in 5 runs, and $R_i = 0$ in 95 runs). Every single promotion made by the system is false ($\mathrm{pFDR} = \mathrm{E}[V/R \mid R > 0] = 1.00$), yet $\bar{X} = 5/100 = \mathbf{0.050} \le \mathbf{0.0776}$, so the gate **passes C2.5 with room to spare**. In cells with zero power (`N2-s3`, where $R_i = 0$ in all 100 runs), $\bar{X} = 0$ trivially, providing zero information about whether promotions are real or noise.
  4. **Missing union bound across mixed cells (internal contradiction with C9.5):** Appendix S4 correctly notes that testing M mixed cells in C9.5 requires a union-bound penalty $\log(M/0.05)/(2n)$, yet applies the unadjusted single-cell penalty $\log(20)/200 = 0.1224$ in C2.5 across multiple generators and planted-Sharpe cells (e.g., for $M = 9$ cells, the 95% family-wise Hoeffding bound requires $\bar{X} \le 0.20 - \log(180)/200 = \mathbf{0.0389}$, less than half of 0.0776).
- **Which Result or Decision It Changes:** It invalidates the claim that passing C2.5 in Stage 2 certifies false-discovery control among promoted strategies when C2.6 failed with 0%–5% promotion rates, because any gate with promotion rate $\le 7.7\%$ passes C2.5 even if 100% of its promotions are false discoveries.

Finding 7 — The Deflated Sharpe Ratio Formula Is Non-Monotone in $\hat{S}$, Has a Finite Ceiling $\Phi(2 \quad T-1/ \quad \hat{\gamma}_4 - 1)$ as $\hat{S} \to \infty$, and Treats Step 2's Multi-N Max as Non-Trivial When N_{raw} Dominates Deterministically

- **Severity:** High
- **Location:** Section 7 (`\label{sec:gate}`, Step 2), Appendix S2 (`\label{app:dsr}`), and Appendix S5 (`\label{app:selection}`):

  > "2. **Screen.** Candidates must pass a deflated Sharpe threshold computed at each of several effective-N estimates, using the largest." "Then $\mathrm{DSR} = \Phi\left(\frac{(\hat{S}-S_0)\quad T-1}{1-\hat{\gamma}_3\hat{S}+(\hat{\gamma}_4-1)\hat{S}^2/4}\right), \quad S_0 = \frac{(1-\gamma_E)\Phi^{-1}(1-1/N)+\gamma_E\Phi^{-1}(1-1/(Ne))}{T-1}$." "The raw count of filled trials is the primary N, so that no favorable estimate is chosen. The complete-linkage $|\rho|$ cuts 0.9 and 0.5 are chosen contrasts for testing fragility, not bounds on the effective number of trials."

- **The Claim:** (a) Evaluating DSR with the plug-in variance denominator $1 - \hat{\gamma}_3\hat{S} + \frac{\hat{\gamma}_4 - 1}{4}\hat{S}^2$ properly penalizes non-normal returns while screening for high Sharpe ratios at $\mathrm{DSR} \ge 0.95$; and (b) computing the DSR threshold at several effective-N estimates ($N_{\mathrm{raw}}, N_{|\rho|=0.9}, N_{|\rho|=0.5}$) and "using the largest" combines multiple estimators.
- **Why It Is Wrong (with Derivation and Counterexample):**
  1. **Non-monotonicity in $\hat{S}$ (higher Sharpe ratios receive lower DSR scores):** Because Appendix S2 plugs the **observed sample Sharpe** $\hat{S}$ (rather than the null benchmark S_0) into the variance denominator $V(\hat{S}) = 1 - \hat{\gamma}_3\hat{S} + c\hat{S}^2$ (where $c = (\hat{\gamma}_4 - 1)/4$), differentiating the DSR z-score $Z(\hat{S}) = \frac{(\hat{S}-S_0)\quad T-1}{V(\hat{S})}$ with respect to $\hat{S}$ gives:
  $$\frac{\partial Z}{\partial \hat{S}} = \quad T-1\,\frac{\left(1 - \frac{1}{2}\hat{\gamma}_3 S_0\right) + \left(cS_0 - \frac{1}{2}\hat{\gamma}_3\right)\hat{S}}{\left(1 - \hat{\gamma}_3\hat{S} + c\hat{S}^2\right)^{3/2}}.$$
  Whenever $\hat{\gamma}_3 > 2cS_0 = \frac{\hat{\gamma}_4 - 1}{2}S_0$ (which holds for **every** positively skewed strategy at $N = 1$ where $S_0 = 0$, and whenever $S_0 < \frac{2\hat{\gamma}_3}{\hat{\gamma}_4 - 1}$), $\frac{\partial Z}{\partial \hat{S}} < 0$ for all $\hat{S} > \hat{S}_{\mathrm{peak}} = \frac{2 - \hat{\gamma}_3 S_0}{\hat{\gamma}_3 - 2cS_0}$: **increasing a strategy's Sharpe ratio $\hat{S}$ strictly lowers its DSR!**

2. **Finite upper bound as $\hat{S} \to \infty$ rejects sparse/episodic strategies regardless of Sharpe:** As $\hat{S} \to \infty$, $Z(\hat{S})$ does not grow without bound; the $\hat{S}$ terms in the numerator and denominator cancel:

$$\lim_{\hat{S}\to\infty} Z(\hat{S}) = \frac{2 \quad T-1}{\hat{\gamma}_4 - 1}.$$

Whenever a strategy has sample kurtosis $\hat{\gamma}_4 > 1 + \frac{4(T-1)}{z_{0.95}^2} \approx 1 + 1.4785(T-1)$ (for instance, an episodic event or extreme-funding strategy that triggers on a single spike or has heavy outlier bars), $\sup_{\hat{S}>S_0} \mathrm{DSR}(\hat{S}) < 0.95$, so **no positive Sharpe ratio can pass the screen**.

3. **Deterministic vacuity of $\max(N_{\mathrm{raw}}, N_{0.9}, N_{0.5})$:** Complete-linkage clustering partitions the N_{raw} filled strategies into N_{clust} clusters, so $1 \le N_{0.5} \le N_{0.9} \le N_{\mathrm{raw}}$ holds deterministically on every dataset. Since $S_0(N)$ is strictly increasing in N, "computed at each of several effective-N estimates, using the largest" in Section 7 Step 2 is algebraically identical to evaluating only N_{raw} and ignoring the cluster estimators entirely.

- **Which Result or Decision It Changes:** It shows that the DSR screen in Step 2 penalizes positively skewed strategies for having *too high* a Sharpe ratio, places an absolute kurtosis ceiling beyond which no Sharpe can pass, and renders the multi-N maximum in Step 2 vacuous.

Finding 8 — N2's Common-Factor t_5 Construction Reduces Innovation Kurtosis by Up to 33.3% (Violating C2.2's 20% Tolerance) and Leaves $\mathrm{GARCH}(1,1)$ Fourth Moments Infinite Before Clipping

- **Severity:** High
- **Location:** Section 12 (`\label{sec:findings}`, "Calibration and sensitivity (stage 2)") and Appendix D1 (`\label{app:nulls}`):

> "N2's GARCH and t_5 with common factor… Variance targeting uses $\omega = s^2(1-\alpha-\beta)$, *derived* from stationary GARCH variance; it keeps near-integrated fits from overstating that target. For unit-variance independent factors $x_i = \lambda z_m + \quad 1-\lambda^2 z_i$, correlation is λ^2, motivating the square root of fitted mean correlation… C2.2's 100-year moment tolerances (5% mean absolute innovation, 20% kurtosis, 0.05 autocorrelations/correlations) are chosen stochastic diagnostics."

- **The Claim:** Constructing correlated innovations via $x_i = \lambda z_m + \quad 1-\lambda^2 z_i$ from independent unit-variance t_5 variables z_m, z_i yields a valid t_5 GARCH process whose moments match the target within C2.2's 20% kurtosis tolerance.
- **Why It Is Wrong (with Recomputation):**

1. **Sum of independent t_5 variables is not t_5 and loses up to 33.3% of its kurtosis:** A standardized unit-variance t_ν variable with $\nu = 5$ has theoretical fourth moment (ordinary kurtosis) $\kappa_{t_5} = 3 + \frac{6}{\nu-4} = 9$. However, the student-$t$ family is not closed under linear combinations. For independent unit-variance $z_m, z_i \sim t_5$, the fourth moment of $x_i = \lambda z_m + \quad 1-\lambda^2 z_i$ is:

$$\mathrm{E}[x_i^4] = \lambda^4 \mathrm{E}[z_m^4] + 6\lambda^2(1-\lambda^2)\mathrm{E}[z_m^2]\mathrm{E}[z_i^2] + (1-\lambda^2)^2\mathrm{E}[z_i^4] = 9\left(\lambda^4 + (1-\lambda^2)^2\right) + 6\lambda^2(1-\lambda^2) = 9 - 12\lambda^2(1-\lambda^2$$

For a cross-coin correlation of $\rho = \lambda^2 = 0.50$ (typical for crypto perpetuals), $\mathrm{E}[x_i^4] = 9 - 12(0.25) = \mathbf{6.0}$, which is 33.3% **below** the t_5 kurtosis of 9.0—violating C2.2's 20% kurtosis tolerance even before GARCH filtering!

2. **Infinite unconditional fourth moment in $\mathrm{GARCH}(1,1)$:** For $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$ with $\varepsilon_t = \sigma_t x_t$, finite unconditional kurtosis requires $(\alpha+\beta)^2 + (\mathrm{E}[x_i^4]-1)\alpha^2 < 1$ (Bollerslev, 1986). For $\mathrm{E}[x_i^4] \in [6, 9]$ and near-integrated crypto GARCH fits ($\alpha+\beta \approx 0.95, \alpha \ge 0.10$), $(\alpha+\beta)^2 + (\mathrm{E}[x_i^4]-1)\alpha^2 \ge 0.9025 + 5(0.01) = 0.9525$ and exceeds 1 whenever $\alpha \ge 0.14$, so the simulated kurtosis in C2.2 is an artifact of the ad-hoc volatility cap 0.0625^2 and return clip $|\varepsilon| \le 0.3$, not the GARCH$(1,1)$-t_5 model.

- **Which Result or Decision It Changes:** It explains why C2.2's generator comparison initially failed on "variance and standardization defects" and shows that N2 does not generate t_5-distributed innovations when $\lambda > 0$.

CONTINUE FOR PART 3

User prompt: continue

Response: Finding 9 — The Unregularized Plug-In Betting Rule $\lambda_t = \min(1/2, \max(0, S_{t-1}/Q_{t-1}))$ Has an Infinite Discontinuity at $r_1 = 0$ and Degenerates into a Bang-Bang $\{0, 1/2\}$ Switch

- **Severity:** High
- **Location:** Section 7 (`\label{sec:gate}`, "The holdout test") and Appendix S3 (`\label{app:bet}`):

"The bet is $\lambda_1 = 0$, then $\min(1/2, \max(0, S_{t-1}/Q_{t-1}))$, zero if $Q = 0$, with $S = \sum r$, $Q = \sum r^2$. Maximizing the second-order approximation $\mathrm{E}\log(1 + \lambda r) \simeq \lambda\mu - \lambda^2\mathrm{E}r^2/2$ gives $\mu/\mathrm{E}r^2$, which motivates the plug-in. The cap $1/2 = 1/L$ is derived, not chosen... Only the plug-in form inside $[0, 1/2]$ is a choice."

- **The Claim:** Using the unregularized sample ratio $S_{t-1}/Q_{t-1} = \left(\sum_{s=1}^{t-1} r_s\right) / \left(\sum_{s=1}^{t-1} r_s^2\right)$ inside $[0, 1/2]$ provides a well-behaved plug-in Kelly fraction approximating $\mu/\mathrm{E}[r^2]$ on the holdout.
- **Why It Is Wrong (with Derivation and Counterexample):**
  1. **Infinite discontinuity at $t = 2$ ($S_1/Q_1 = 1/r_1 \to +\infty$ as $r_1 \to 0^+$):** After step 1 ($\lambda_1 = 0$), the estimator at step 2 is:
  $$\frac{S_1}{Q_1} = \frac{r_1}{r_1^2} = \frac{1}{r_1}.$$
  Unlike regularized betting martingales (such as Waudby-Smith and Ramdas 2024, which add a prior pseudo-variance $c > 0$ to the denominator or shrink λ_t at rate $O(1/\quad t)$), $S_1/Q_1 = 1/r_1$ explodes as $r_1 \to 0^+$. For **every** positive first-bar return $r_1 \in (0, 2]$—even a microscopic floating-point dust gain of $r_1 = +10^{-12}$—the ratio $1/r_1 \geq 1/2$ immediately slams λ_2 to the maximum possible cap $\lambda_2 = 1/2$, whereas at $r_1 = 0$ (or $r_1 = -10^{-12}$), $\lambda_2 = 0$. *Counterexample:* If a strategy has $r_1 = +10^{-6}$ on bar 1 and experiences a single large loss $r_2 = -1.80$ on bar 2, $\lambda_2 = 1/2$ causes the wealth process to drop to $K_2 = 1 + \frac{1}{2}(-1.80) = 0.10$ (a **90% wealth collapse on bar 2** triggered by a 10^{-6} noise tick on bar 1), requiring a $200\times$ gain thereafter to reach $K_t = 20$.
  2. **Bang-bang saturation across the entire holdout:** At step t, λ_t hits the upper cap $1/2$ whenever $\bar{r}_{t-1} \geq \frac{1}{2}\overline{r^2}_{t-1} \approx \frac{1}{2}\sigma^2$, and hits the lower cap 0 whenever $\bar{r}_{t-1} \leq 0$. In terms of the running t-statistic $Z_{t-1} = \frac{\bar{r}_{t-1}}{\sigma/\ \ t-1}$, the interior region $\lambda_t \in (0, 1/2)$ is only:
  $$0 < Z_{t-1} < \frac{1}{2}\sigma \quad t - 1.$$
  At the paper's baseline volatility $\sigma = 0.02$ per 4-hour bar and holdout length $n = 546$ ($\quad 545 = 23.35$), the upper boundary of the interior region is $Z_{t-1} = 0.01 \quad t - 1 \leq \mathbf{0.2335}$. Under H_0, $P(0 < Z_{t-1} < 0.01 \quad t - 1) = \Phi(0.01 \quad t - 1) - 0.5$ is at most 9.2% at $t = 546$ (and $< 1.3\%$ for $t \leq 10$). Over more than 90% of the distribution, S_{t-1}/Q_{t-1} never acts as a Kelly fraction inside $(0, 1/2)$; it acts as a discontinuous bang-bang switch $\lambda_t \in \{0, 1/2\}$.
- **Which Result or Decision It Changes:** It invalidates the second-order interior Kelly motivation in Appendix S3, explains why early-step rare losses caused the holdout calibration suites (v1 and v2.1) to fail or lose power, and makes the holdout p-value hypersensitive to the sign of the very first active bar.

Finding 10 — Locked Holdout Suite v3's Slack Tolerances ($0.0625, 0.015, 0.01$) Pass Broken Implementations with Up to 60%–80% Inflated Type I Error

- **Severity:** High
- **Location:** Section 12 (`\label{sec:findings}`, "Holdout-test studies") and Appendix S4 (`\label{app:calibration}`):

  "The human adopted the bounded-return betting test, and its fresh v3 suite passed its 99 judged pairs. That checks the implementation on bounded conditional-null series..." "The locked holdout suites choose 20,000 series per cell, lengths 540, 546 and 1,080, and levels 0.05, 0.01 and 0.005. Bonferroni confidence $1 - 0.05/K$ is derived to cover all K pairs (54 in v1 and v2, 99 in v3). The tolerances 0.0625, 0.015 and 0.01 are chosen slack for an implementation check."
- **The Claim:** Requiring the $1 - 0.05/99$ one-sided upper confidence bound across $n = 20,000$ simulated null series per cell to fall below 0.0625 (at $\alpha = 0.05$), 0.015 (at $\alpha = 0.01$), and 0.010 (at $\alpha = 0.005$) verifies the holdout test implementation.
- **Why It Is Wrong (with Recomputation):**
  1. For v3's exact Ville supermartingale test ($p = \min(1, 1/\max_t K_t)$), the theoretical rejection rate under any valid bounded conditional-null series is strictly bounded by α (**not** merely asymptotically $\approx \alpha$, and typically $\ll \alpha$ due to supermartingale overshoot at the stopping boundary).
  2. With $n = 20,000$ series and $K = 99$ pairs ($\delta_{\text{Bonf}} = 0.05/99 = 5.05 \times 10^{-4}$, $z_{1-\delta} = 3.2877$), if an implementation actually satisfies $p_{\text{true}} \leq \alpha$, the expected upper Clopper–Pearson bound at $\alpha = 0.05, 0.01, 0.005$ is at most:
     - At $\alpha = 0.05$: $q_{\text{upper}}(1000; 20000) = \mathbf{0.0553}$ (whereas the tolerance is 0.0625).
     - At $\alpha = 0.01$: $q_{\text{upper}}(200; 20000) = \mathbf{0.0125}$ (whereas the tolerance is 0.0150).
     - At $\alpha = 0.005$: $q_{\text{upper}}(100; 20000) = \mathbf{0.0068}$ (whereas the tolerance is 0.0100).

3. Because v3 kept the wide heuristic tolerances $(0.0625, 0.015, 0.010)$ from the failed asymptotic tests (v1/v2), a bugged implementation whose true false-rejection rate at $\alpha = 0.005$ is actually $p_{\text{true}} = 0.0080$ (a $+60\%$ **relative inflation** in Type I error, which is $+6.0\sigma$ above $\alpha = 0.005$ at $n = 20,000$) produces $x = 160$ rejections with a $1 - 0.05/99$ Clopper–Pearson upper bound of:

$$q_{\text{upper}}(160; 20000) = \mathbf{0.00999} < \mathbf{0.0100},$$

and therefore **passes v3**! Similarly, at $\alpha = 0.01$ (the exact Holm threshold for $k = 5$ finalists in Stage 9!), an implementation with $p_{\text{true}} = 0.0124$ ($+3.4\sigma$ above 0.01) has $q_{\text{upper}}(248; 20000) = 0.01498 < 0.0150$ and passes v3.

- **Which Result or Decision It Changes:** It shows that v3's tolerances are mis-specified for an exact non-asymptotic supermartingale test and fail to reject implementation bugs that inflate tail false-discovery rates by 24%–60% at the Holm-adjusted levels ($\alpha = 0.01$ and 0.005).

Finding 11 — Stationary Bootstrap Block Length ($b = 6$ Four-Hour Bars = 24 Hours) Is $7\times$ Shorter Than the Null Generator's Own 168-Hour Funding Blocks

- **Severity:** Medium
- **Location:** Section 7 (`\label{sec:gate}`, Table 3 & Steps 3–4), Appendix S5 (`\label{app:selection}`), and Appendix D1 (`\label{app:nulls}`):

  > "White's Reality Check, Hansen's SPA… Key assumption to check: A correctly specified family and a suitable bootstrap for dependent data." "DSR 0.95, SPA/StepM 0.05, stationary-bootstrap mean block 6 four-hour steps and 1,000 resamples are *choices*: a stringent screen and a reproducible, affordable heuristic for dependence." "Funding and premiums are resampled jointly in stationary blocks of mean length 168 hours, which retains their observed floor, tails and cross-coin dependence."

- **The Claim:** A stationary bootstrap with geometric mean block length $b = 6$ four-hour steps (24 hours) adequately captures serial dependence for Hansen's SPA and Romano–Wolf StepM in the pilot's funding-dislocation domain.
- **Why It Is Wrong (with Recomputation):**

  1. In the pilot's target strategy family (funding dislocations, Stage 0 and Appendix D2), positions are driven by persistent funding rates and premiums whose autocorrelation structure spans a full week—which is why Appendix D1 explicitly requires a stationary block length of $168 \text{ hours} = 42$ **four-hour bars** to preserve their serial dependence!

  2. When a strategy's trading signal or conditional variance has geometric persistence at the 42-bar (168h) scale (with lag-k autocorrelation $\rho_k = \phi^k$, where $\phi = 1 - 1/42 = 0.9762$), the true long-run variance factor relative to i.i.d. variance is $\frac{1+\phi}{1-\phi} = 83.0$. Resampling with mean block length $b = 6$ (continuation probability $p_{\text{boot}} = 1 - 1/6 = 0.8333$) attenuates lag-$k$ autocovariances by $(p_{\text{boot}})^k = (5/6)^k$, capturing only:

  $$\frac{1 + \phi p_{\text{boot}}}{1 - \phi p_{\text{boot}}} = \frac{1 + 0.9762(5/6)}{1 - 0.9762(5/6)} = \mathbf{9.73} \quad \text{vs. } \mathbf{83.0},$$

  under-estimating the long-run variance of persistent series by up to $8.5\times$ (or under-estimating standard errors by $2.9\times$), which inflates studentized t-statistics in SPA and Romano–Wolf.

- **Which Result or Decision It Changes:** It invalidates the bootstrap dependence assumption in Table 3 for any strategy whose returns or variance inherit the 168-hour persistence built into the pilot's own funding generators.

Finding 12 — Section 10's 4-Pattern Attack Suite Cannot Certify the Licensing Boundary Against 3-Query Linear Reconstruction ($r_{i,t^*} = \frac{1}{2}(|C_1|\mu_1 + |C_2|\mu_2 - |C_3|\mu_3)$)

- **Severity:** Medium
- **Location:** Section 10 (`\label{sec:multiasset}`, "Licensing boundary") and Appendix D3 (`\label{app:scope}`):

  > "What may cross: the fields of a frozen per-source report list, each with its minimum aggregation stated as a count of decision dates and instruments per reported cell: for example results by year and by regime, contribution by instrument and class, turnover, exposure, hit rate, results split by the strategy's own condition… The boundary belongs to the disclosure path (Section 6), and certification runs a frozen attack suite against it: single-date conditions, one-date differences between two strategies, long-only single-instrument strategies, and threshold bisection on a feature. It passes only if no per-date value is recovered within a stated tolerance from any combination of reports within a campaign's evaluation budget."

- **The Claim:** Enforcing a minimum aggregation count m per reported cell, forbidding calendar-date conditions, and passing the four-pattern attack suite (single-date conditions, one-date differences between two strategies, long-only single-instrument strategies, and threshold bisection) certifies that no per-date value can be recovered from any combination of reports within a campaign's evaluation budget.

- **Why It Is Wrong (with Concrete Counterexample):** Because the disclosure path returns exact (un-noised) conditional cell averages $\mu(C) = \frac{1}{|C|}\sum_{(i,t)\in C} r_{i,t}$ and cell counts $|C|$ split by the strategy's own endogenous regime condition C and by instrument, preventing pairwise 1-date diffs ($|C_1 \triangle C_2| = 1$) does **not** prevent 3-way linear reconstruction:
  - Let A and B be two disjoint sets of m dates defined by non-calendar feature ranges (e.g., volatility deciles), and let $t^* \notin A \cup B$ be the target date where a secondary feature y_t attains its maximum.
  - Submit three strategies whose regime conditions select $C_1 = A \cup \{t^*\}$ (size $m+1$), $C_2 = B \cup \{t^*\}$ (size $m+1$), and $C_3 = A \cup B$ (size $2m$), each trading a multi-instrument basket (since "contribution by instrument" is disclosed separately).
  - Every cell and its complement has size $\geq m \gg 1$, no strategy is single-instrument, no strategy uses calendar dates or threshold bisection, and every pair of strategies differs on at least $m+1$ dates ($|C_1 \triangle C_2| = 2m$, $|C_1 \triangle C_3| = m+1$, $|C_2 \triangle C_3| = m+1$), so all four attack filters pass.
  - Yet the exact single-date, single-instrument return r_{i,t^*} is recovered deterministically in just **3 evaluations**:

$$r_{i,t^*} = \frac{1}{2}\Big((m+1)\mu_i(C_1) + (m+1)\mu_i(C_2) - 2m\mu_i(C_3)\Big).$$

- **Which Result or Decision It Changes:** It proves that without differential-privacy noise on disclosed aggregates or restricting regime splits to a fixed strategy-independent partition, the licensing boundary's certification criterion ("no per-date value is recovered… from any combination of reports") is mathematically violated by 3-query tracker attacks.

Finding 13 — Table 1's Noise Ceiling $h(N,T) = \quad 2\log N/T$ Overstates the Exact Gaussian Expected Maximum by 39.5% at $N = 10$

- **Severity:** Medium
- **Location:** Section 2.1 (`\label{sec:budget}`, Table 1) and Appendix S1 (`\label{app:noise}`):

  > "For independent $Z_i \sim N(0,1)$, $\E \max_i \Shat_i = \SE \E \max_i Z_i$, $\E \max_i Z_i \sim \quad 2\log N$, $h(N,T) = \quad 2\log N/T$."

- **The Claim:** Table 1 reports the approximate noise ceiling $h(N,T)$ for $N \in \{10, 1000, 100000\}$ and $T \in \{3, 10, 20\}$ years as an approximation to $\E[\max_{1\leq i\leq N} \hat{S}_i]$.
- **Why It Is Wrong (with Recomputation):** Because the Gumbel convergence $\E[\max_{1\leq i\leq N} Z_i] / \quad 2\log N \to 1$ has an $O(\frac{\log\log N}{\log N})$ negative correction, using the leading term $\quad 2\log N$ at small N severely overstates the expected maximum:
  - At $N = 10$, exact numerical integration gives $\E[\max_{1\leq i\leq 10} Z_i] = \mathbf{1.5388}$, whereas $\quad 2\log 10 = \mathbf{2.1460}$ (a $+39.5\%$ **overestimate**). The true expected maxima at $T = 3, 10, 20$ years are $0.89, 0.49, 0.34$—not Table 1's $1.24, 0.68, 0.48$.
  - At $N = 1,000$, $\E[\max_{1\leq i\leq 1000} Z_i] = \mathbf{3.2414}$ vs. $\quad 2\log 1000 = \mathbf{3.7169}$ ($+14.7\%$ overestimate): the true values at $T = 3, 10, 20$ years are $1.87, 1.02, 0.72$—not Table 1's $2.15, 1.18, 0.83$.
- **Which Result or Decision It Changes:** It inflates every entry of Table 1 (`\label{tab:ceiling}`), particularly the $N = 10$ row (which represents the confirmation family size in Section 9), by up to 39.5%.

Finding 14 — Stage 1's Canary Martingale Leak Threshold (Annualized Sharpe ≥ 4 Over 550 Days, or 4.91σ) Is Blind to Subtle Causal Leaks

- **Severity:** Low
- **Location:** Section 12 (`\label{sec:findings}`, "Kernel MVP (stage 1)") and Appendix K1 (`\label{app:kernel}`):

  > "C1.7's Gaussian log-return mean must be $-\sigma^2/2$, *derived* from $\E e^X = e^{\mu+\sigma^2/2}$ so that prices are a martingale. The chosen 2% volatility, 550 days, admitted and leaking Sharpes of 4 and 20, six fixed and five fresh seeds and 20 random programs have no joint calibration of false alarms and detection."

- **The Claim:** Running 20 random programs across 11 seeds on a 550-day ($T_{\text{yr}} = 550/365 = 1.5068$ yr) martingale canary market with an admitted annualized Sharpe cutoff of 4.0 tests for causal leakage in the evaluator.
- **Why It Is Wrong (with Recomputation):** Over $T_{\text{yr}} = 1.5068$ years, the standard error of an annualized Sharpe ratio under the martingale null is $\SE(\hat{S}) = T_{\text{yr}}^{-1/2} = \quad 365/550 = \mathbf{0.8146}$. An admitted threshold of $\hat{S} = 4.0$ corresponds to a single-run z-score of $4.0/0.8146 = \mathbf{4.91}\sigma$ ($p = 4.5 \times 10^{-7}$). Any partial or conditional lookahead bug (e.g., a 1-bar intrabar close/fill leak on a sparse signal or

a subtle window-boundary off-by-one) that generates a true annualized Sharpe of $S_{\text{true}} = 2.0$ ($z_{\text{mean}} = 2.455\sigma$) will fall below the 4.0 alarm threshold on $1 - \Phi((4.0 - 2.0)/0.8146) = \mathbf{99.3\%}$ of individual runs.

- **Which Result or Decision It Changes:** It shows that passing C1.7 only rules out full-bar clairvoyance ($\hat{S} \geq 4$) and cannot detect moderate lookahead bugs ($\hat{S} \in [1.5, 3.0]$).
